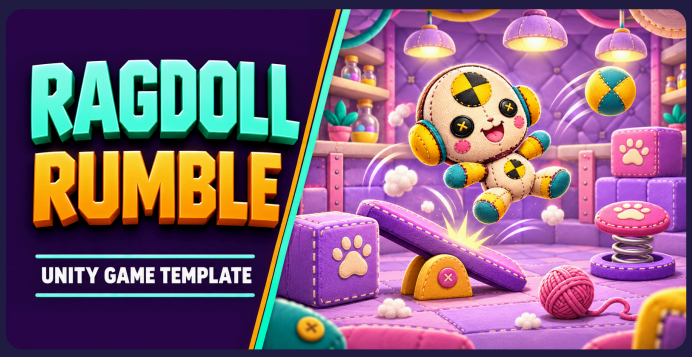
UNITY 6000.0.62F1 · COMPLETE OFFLINE DOCUMENTATION

Ragdoll Rumble Developer Guide

Install. Edit. Reskin. Build your release.

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CHAPTER 01

Start here: Ragdoll Rumble

Ragdoll Rumble is a standalone editable Unity game template. Drag and fling the cartoon Stuffies around the physics arena. Empty their stuffing bars and complete the displayed arena goal. Use the actual tool targeting actions and interact with the supplied props and hazards.

Includes 25 curated levels, source-backed generation, a game-specific visual editor and validation, 4 guided helpers, local saving and portrait/landscape scenes. Keep the required private game Kit and publisher-owned Szekipapa77Ads layer. Provider SDK binaries are excluded. There is no bundled HLK, purchases, multiplayer server or remote LiveOps backend. Import only one game Kit into a project because the Kit classes share names.

CHAPTER 02

Requirements, installation and first play

Use Unity **6000.0.62f1**, URP **17.0.4** and Unity UI **2.0.0**. Install matching Android SDK/NDK/OpenJDK and WebGL modules in Unity Hub. Retain the supplied manifest and ProjectSettings.

1. Extract the standalone project to a short writable local path, then open it in the matching editor. Wait for package import and compilation.
2. For native unitypackage import, create a matching URP project and use Assets > Import Package > Custom Package. Import every game, Kit, Szekipapa77Ads and WebGL-template entry.
3. Run Szekipapa77 > Ragdoll Rumble > Setup. It configures rendering and player settings; review its changes when integrating with existing work.
4. Active Input Handling must support the classic Input Manager: choose Old or Both. New Input System only is insufficient.
5. Open `Assets/Games/RagdollRumble/Scenes/RagdollRumble_Mobile.unity` and press Play. Follow the level card and first-level instructions.
6. Stop Play mode before editing level assets. Keep all .meta GUIDs. The game uses classic Unity UI Text; do not assume an extra TextMesh Pro import is required.

CHAPTER 03

Project folders and exact source map

All game-relative paths below start at `Assets/Games/RagdollRumble`. Runtime UI is built in code; editing an unrelated static Canvas does not reskin it. Scripts contains game rules and views; Editor contains authoring tools; Kit supplies UI/audio/save/platform support; Resources/ragdollrumble contains level JSON, art, music and sound; Settings owns URP assets; Scenes contains both entry scenes.

- `Scripts/RagdollRumbleGame.cs`
- `Scripts/RagdollRumbleGame.Hud.cs`
- `Scripts/RagdollRumbleGame.View.cs`
- `Scripts/RrArt.cs`
- `Scripts/RrBot.cs`
- `Scripts/RrData.cs`
- `Scripts/RrGen.cs`
- `Scripts/RrHold.cs`
- `Scripts/RrLevels.cs`
- `Scripts/RrSfx.cs`
- `Scripts/RrSim.cs`

Main flow: RagdollRumbleGame.cs. UI and popup consumers: RagdollRumbleGame.Hud.cs. Model/generator/loading: RrData.cs, RrGen.cs, RrLevels.cs, RrSim.cs. Authoring: Editor/RagdollRumbleLevelEditor.cs. Shared ad settings live outside the game at Assets/Szekipapa77Ads/Resources/AdsSettings.asset.

CHAPTER 04

Gameplay, controls and progression

Drag and fling the cartoon Stuffies around the physics arena. Empty their stuffing bars and complete the displayed arena goal. Use the actual tool targeting actions and interact with the supplied props and hazards.

The initial bank has 25 curated levels. Read the supplied loader/generator before changing continuation rules. Play the guided opening, ordinary progress, completion and retry paths; preserve saved indexes during a visual reskin. The Mobile scene is portrait and the WebGL-named scene is landscape, but the supplied Build menu selects portrait for both build targets. Select landscape explicitly in your Build Profile if desired. Test a near-startup scene replacement as well as a normal restart.

CHAPTER 05

Tutorials and every helper

Fresh local progress triggers the first-level guidance. Newly unlocked helpers have introduction and guided-use flows. Reset only a test save when reviewing tutorials. A free guided first use can differ from ordinary charge consumption. Exact helper names/unlocks/charges: Scripts/RagdollRumbleGame.cs; help and effect descriptions: Scripts/RagdollRumbleGame.cs.

Helper	Unlock	Free charges	Action and effect
Punch	3	3	A spring-loaded glove flies in and bonks whatever you tap!
Rocket	6	3	A party rocket bursts where you tap and blasts everything away.
Anvil	10	2	Drops a heavy anvil from the ceiling right above your tap.
Vortex	15	2	A swirling vortex pulls everything in for 2 seconds, then pops!

Activate the tool and perform its required target action: arming a tool alone does not demonstrate its effect. A successful use requires the stated preconditions from this game: show both activation and its actual effect. Exhausted charges may offer a rewarded ad; verify success/cancel/failure and grant a reward once only.

CHAPTER 06

Visual level editor: worked editing session

Open **Szekipapa77 > Ragdoll Rumble > Level Editor**. Exact implementation: `Editor/RagdollRumbleLevelEditor.cs`.

Duplicate an early entry in this game-specific visual editor, change its visible name and one rule, save, validate, then play it. Use the exact source-control table below for this game.

1. Back up Resources/ragdollrumble/Levels/levels.json.
2. Duplicate a simple early level and edit one small element.
3. Use the game-specific validation controls and resolve every reported issue.
4. Save explicitly; stop and restart Play mode, then use Play this level or navigate to the edited index.
5. Check portrait and landscape and rebuild demo outputs after accepting data changes.

The exact current authoring controls, including tooltips and generator switches, follow. These are game-specific source references, not a generic editor feature list.

Source	Exact current code
`RagdollRumbleLevelEditor.cs:46`	`[MenuItem("Szekipapa77/Ragdoll Rumble/Level Editor")]`
`RagdollRumbleLevelEditor.cs:49`	`[MenuItem("Szekipapa77/Ragdoll Rumble/Validate All Levels")]`
`RagdollRumbleLevelEditor.cs:120`	`if (GUILayout.Button(_dirty ? "Save *" : "Save", EditorStyles.toolbarButton, GUILayout.Width(70))) Save();`
`RagdollRumbleLevelEditor.cs:121`	`if (GUILayout.Button("Reload", EditorStyles.toolbarButton, GUILayout.Width(60)) && (!_dirty || EditorUtility.DisplayDialog("Reload", "Discard unsaved changes?", "Discard", "Cancel"))) Load();`
`RagdollRumbleLevelEditor.cs:124`	`if (GUILayout.Button(new GUIContent("Validate", "The bot plays this arena with drags in the real physics; stores plan, moves, par and difficulty"), EditorStyles.toolbarButton, GUILayout.Width(70)) && Cur != null) ValidateOne(_sel);`
`RagdollRumbleLevelEditor.cs:125`	`if (GUILayout.Button("Validate All", EditorStyles.toolbarButton, GUILayout.Width(85))) ValidateAll();`

RagdollRumbleLevelEditor.cs:128	if (GUILayout.Button(new GUIContent("Play this level", "Quick test only: saves, opens the WebGL scene at this level and enters Play mode."), EditorStyles.toolbarButton, GUILayout.Width(100)) && Cur != null) Play(_sel + 1);
RagdollRumbleLevelEditor.cs:139	if (GUILayout.Button("Add")) { var l = RrGen.Generate(_file.levels.Length + 1, 0.3f, Environment.TickCount & 0xFFFF, false); l.Normalize(); Insert(_file.levels.Length, l); }
RagdollRumbleLevelEditor.cs:140	if (GUILayout.Button("Dup") && Cur != null) Insert(_sel + 1, Cur.Clone());
RagdollRumbleLevelEditor.cs:141	if (GUILayout.Button("Del") && Cur != null && EditorUtility.DisplayDialog("Delete", $"Delete level {_sel + 1}?", "Delete", "Cancel"))
RagdollRumbleLevelEditor.cs:146	if (GUILayout.Button("Up") && _sel > 0) Swap(_sel, _sel - 1);
RagdollRumbleLevelEditor.cs:147	if (GUILayout.Button("Down") && _sel < _file.levels.Length - 1) Swap(_sel, _sel + 1);
RagdollRumbleLevelEditor.cs:189	EditorGUILayout.LabelField("Drag handles (purple = part, red = dummy feet). Right-click deletes. 0.1 m snap.", EditorStyles.miniLabel);
RagdollRumbleLevelEditor.cs:191	if (GUILayout.Button("Add part", GUILayout.Width(70))) AddPart(L, PartTypes[_addType]);
RagdollRumbleLevelEditor.cs:193	if (GUILayout.Button("Add dummy", GUILayout.Width(80))) AddDummy(L, RrProfile.All[_addProfile].id);
RagdollRumbleLevelEditor.cs:198	_scrub = EditorGUILayout.Slider("Mover time (s)", _scrub, 0, 8);
RagdollRumbleLevelEditor.cs:417	EditorGUILayout.LabelField($"Level {_sel + 1}", EditorStyles.largeLabel);
RagdollRumbleLevelEditor.cs:419	L.name = EditorGUILayout.TextField("Name", L.name);
RagdollRumbleLevelEditor.cs:420	EditorGUILayout.LabelField("First-time hint");

RagdollRumbleLevelEditor.cs:422	L.mech = EditorGUILayout.TextField(new GUIContent("Introduces", "Mechanic id (spikes, goo, zapper, saw, press, pendulum, seesaw, bumper, fan, pad, crates, cage, moon, two) or a profile id"), L.mech);
RagdollRumbleLevelEditor.cs:424	L.theme = ThemeNames[EditorGUILayout.Popup("Theme", ti, ThemeNames)];
RagdollRumbleLevelEditor.cs:425	L.gravity = EditorGUILayout.Slider("Gravity", L.gravity, 0.3f, 1.5f);
RagdollRumbleLevelEditor.cs:426	L.difficulty = EditorGUILayout.Slider("Design difficulty", L.difficulty, 0, 1);
RagdollRumbleLevelEditor.cs:427	L.breather = EditorGUILayout.Toggle("Breather", L.breather);
RagdollRumbleLevelEditor.cs:428	L.par = EditorGUILayout.IntSlider("Par (3 stars)", L.par, 1, 12);
RagdollRumbleLevelEditor.cs:429	L.seed = EditorGUILayout.IntField("Seed", L.seed);
RagdollRumbleLevelEditor.cs:430	L.recipe = EditorGUILayout.TextField(new GUIContent("Recipe", "Motifs for Recompose: " + string.Join(" ", RrGen.Motifs)), L.recipe);
RagdollRumbleLevelEditor.cs:433	if (GUILayout.Button(new GUIContent("Recompose", "Rebuild parts and dummies from recipe + seed + difficulty (RrGen.Compose), then validate"))) Recompose(L, L.seed);
RagdollRumbleLevelEditor.cs:434	if (GUILayout.Button("New seed")) Recompose(L, UnityEngine.Random.Range(1, 99999));
RagdollRumbleLevelEditor.cs:436	EditorGUILayout.LabelField($"{L.parts.Length} parts, {L.dummies.Length} dummies, bot {L.botMoves} moves, score {L.botScore:0}", EditorStyles.miniLabel);
RagdollRumbleLevelEditor.cs:438	EditorGUILayout.HelpBox($"{(v.ok ? "CLEARED" : "NOT CLEARED")}: {v.note}\nmoves {v.moves}, difficulty {v.difficulty:0}/100, {v.time:0.0}s of play", v.ok ? MessageType.Info : MessageType.Error);

RagdollRumbleLevelEditor.cs:453	EditorGUILayout.LabelField("Selected part", EditorStyles.boldLabel);
RagdollRumbleLevelEditor.cs:454	if (L == null || _selPart < 0 || _selPart >= L.parts.Length) { EditorGUILayout.HelpBox("Click a part handle on the canvas (purple squares).", MessageType.None); return; }
RagdollRumbleLevelEditor.cs:458	p.type = PartTypes[EditorGUILayout.Popup("Type", ti, PartTypes)];
RagdollRumbleLevelEditor.cs:459	p.a = EditorGUILayout.Vector2Field(p.type == "pendulum" ? "Pivot" : "a", p.a);
RagdollRumbleLevelEditor.cs:460	if (RrPart.HasB(p.type)) p.b = EditorGUILayout.Vector2Field("b", p.b);
RagdollRumbleLevelEditor.cs:461	p.w = EditorGUILayout.FloatField(p.type == "ball" || p.type == "pendulum" || p.type == "saw" || p.type == "bumper" ? "Radius" : "Width", p.w);
RagdollRumbleLevelEditor.cs:463	p.h = EditorGUILayout.FloatField(p.type == "pendulum" ? "Rope length" : "Height", p.h);
RagdollRumbleLevelEditor.cs:465	p.rot = EditorGUILayout.FloatField(p.type == "pendulum" ? "Start angle" : p.type == "seesaw" ? "Tilt limit" : "Rotation", p.rot);
RagdollRumbleLevelEditor.cs:466	if (p.type == "zapper" || p.type == "saw" || p.type == "press") { p.period = EditorGUILayout.FloatField("Period (s, 0 = still/always on)", p.period); p.phase = EditorGUILayout.Slider("Phase", p.phase, 0, 1); }
RagdollRumbleLevelEditor.cs:467	if (p.type == "ball" || p.type == "pendulum" || p.type == "fan") p.k = EditorGUILayout.FloatField(p.type == "fan" ? "Force" : "Mass", p.k);
RagdollRumbleLevelEditor.cs:475	EditorGUILayout.LabelField("Selected dummy", EditorStyles.boldLabel);
RagdollRumbleLevelEditor.cs:476	if (L == null || _selDummy < 0 || _selDummy >= L.dummies.Length) { EditorGUILayout.HelpBox("Click a dummy on the canvas.", MessageType.None); return; }

RagdollRumbleLevelEditor.cs:480	`pi = EditorGUILayout.Popup("Profile", pi, RrProfile.All.Select(p => p.name).ToArray());`
RagdollRumbleLevelEditor.cs:482	`d.pos = EditorGUILayout.Vector2Field("Feet", d.pos);`
RagdollRumbleLevelEditor.cs:484	`EditorGUILayout.LabelField($"hp {P.hp:0}, mass x{P.massMul:0.0}, size x{P.scale:0.00}. {P.blurb}", EditorStyles.wordWrappedMiniLabel);`
RagdollRumbleLevelEditor.cs:491	`EditorGUILayout.LabelField("Difficulty curve (bot score; blue = breather, red = not cleared). Click a bar.", EditorStyles.miniLabel);`
RagdollRumbleLevelEditor.cs:510	`_showGen = EditorGUILayout.Foldout(_showGen, "Auto Generate (same RrGen code as the runtime)", true);`
RagdollRumbleLevelEditor.cs:512	`_genCount = EditorGUILayout.IntField("Count", _genCount);`
RagdollRumbleLevelEditor.cs:513	`_genStart = EditorGUILayout.IntField("Start level #", _genStart);`
RagdollRumbleLevelEditor.cs:514	`_genSeed = EditorGUILayout.IntField("Seed", _genSeed);`
RagdollRumbleLevelEditor.cs:515	`_genFrom = EditorGUILayout.Slider("Start difficulty", _genFrom, 0, 1);`
RagdollRumbleLevelEditor.cs:516	`_genTo = EditorGUILayout.Slider("End difficulty", _genTo, 0, 1);`
RagdollRumbleLevelEditor.cs:517	`_genBreather = EditorGUILayout.IntSlider("Breather every", _genBreather, 0, 8);`
RagdollRumbleLevelEditor.cs:518	`_genReplace = EditorGUILayout.ToggleLeft("Replace the range starting at 'Start level #' (otherwise insert)", _genReplace);`
RagdollRumbleLevelEditor.cs:528	`if (GUILayout.Button($"Generate {_genCount} validated arenas")) Generate();`
RagdollRumbleLevelEditor.cs:560	`_log = $"Generated {made.Count} validated arenas at #{at + 1} ({(_genReplace ? "replaced" : "inserted")}).";`

CHAPTER 07

Generator, insertion and replacement

Use this editor generation panel, choose a reproducible seed and small count, review start/index and replace behavior, then generate, save and reopen. The source excerpt below identifies exact append/replace semantics.

Source	Exact current code
`RagdollRumbleLevelEditor.cs:42`	`bool _genReplace, _showGen = true;`
`RagdollRumbleLevelEditor.cs:518`	`_genReplace = EditorGUILayout.ToggleLeft("Replace the range starting at 'Start level #' (otherwise insert)", _genReplace);`
`RagdollRumbleLevelEditor.cs:556`	`if (_genReplace) { for (int i = 0; i < made.Count; i++) { if (at + i < list.Count) list[at + i] = made[i]; else list.Add(made[i]); } }`
`RagdollRumbleLevelEditor.cs:557`	`else list.InsertRange(at, made);`
`RagdollRumbleLevelEditor.cs:560`	`_log = $"Generated {made.Count} validated arenas at #{at + 1} ({(_genReplace ? "replaced" : "inserted")}).";`

Worked extension: back up the bank, generate a small set at moderate difficulty using the exact per-game controls, validate them, then Save. Reopen the JSON and confirm the count changed. Do not assume every editor has the same insertion behavior: the actual implementation below distinguishes insert, append and replace. Runtime continuation does not automatically append new records to project JSON. Validation and bot scores help find problems but do not certify a pleasant human difficulty curve.

CHAPTER 08

Actual level JSON and identifiers

Canonical asset: `Assets/Games/RagdollRumble/Resources/ragdollrumble/Levels/levels.json`. Array order controls visible numbering. Preserve numeric/boolean types, root version fields where present and resource/save identifiers. The actual first record is shown below; use the editor for structured coordinates and block/word data.


```
{
  "name": "First Fling",
  "hint": "Drag the Stuffy and fling it. Hard hits empty its stuffing!",
  "mech": "",
  "theme": "lab",
  "recipe": "tower",
  "seed": 1003,
  "difficulty": 0.10000000149011612,
  "breather": false,
  "gravity": 1.0,
  "dummies": [
    {
      "pos": {
        "x": -4.239999771118164,
        "y": -1.590000033378601
      },
      "profile": "buddy"
    }
  ],
  "parts": [
    {
      "type": "block",
      "a": {
        "x": -4.239999771118164,
        "y": -3.5399999618530273
      },
      "b": {
        "x": 0.0,
        "y": 0.0
      },
      "w": 1.7999999523162842,
      "h": 3.9100000858306885,
      "rot": 0.0,
      "period": 0.0,
      "phase": 0.0,
      "k": 0.0
    }
  ],
  "plan": "24,0,0,0,0,0,1.26,-0.23,9;",
  "botMoves": 1,
  "botScore": 5.0,
  "par": 2
}
```


Root keys: levels. Inspect RrData.cs, RrGen.cs, RrLevels.cs, RrSim.cs for field semantics and validation. Keep visible names/hints distinct from technical keys. The game-specific board, wave, card, opponent or question records are gameplay data; UI translation is separate.

CHAPTER 09

Reskin recipe: interface colours

Stop Play mode and open `Scripts/RrArt.cs`. Find `Cream`, currently `#FFF4E0`. Change that hex literal to `#42C9B0`, preserve its symbol/type, compile and reopen the scene. Review HUD, menus, helper cards, settings/results and both orientations. Art sprites and procedural fallback shapes can have additional colours; edit them deliberately.

Actual consumers of this symbol are listed first, then all current palette/hex assignments. Editor-only palettes do not control the runtime. Preserve state/goal contrast and rebuild after accepting a change.

Source	Exact current code
`RagdollRumbleGame.View.cs:59 / void BuildViewRoot()`	`tex.SetPixel(0, 1, RrArt.Lilac); tex.SetPixel(0, 0, RrArt.Cream); tex.Apply();`
`RagdollRumbleGame.View.cs:415 / void BuildPropView(RrSim.Prop pr)`	`mb.Circle(Vector2.zero, r * 0.72f, RrArt.Sun, 24); mb.Circle(Vector2.zero, r * 0.45f, RrArt.Cream, 20);`
`RagdollRumbleGame.View.cs:818 / void StuffingFx(Vector2 at, int n)`	`for (int i = 0; i < n; i++) Spawn(at, Random.insideUnitCircle * 4f + Vector2.up * 2f, i % 3 == 0 ? RrArt.Cream : Color.white, 0, Random.Range(0.18f, 0.34f), Random.Range(0.5f, 0.9f));`

Source	Exact current code
`RagdollRumbleGame.cs:27 / class/static declarations`	`public static readonly string[] ToolColor = { "#FF6FA5", "#2EC4B6", "#8E7CC3", "#FF8A3D" };`
`RagdollRumbleGame.View.cs:39 / class/static declarations`	`public static readonly Color[] BlockColors = { HzUI.Hex("#C9B3EC"), HzUI.Hex("#FFB8D2"), HzUI.Hex("#A8E6DF"), HzUI.Hex("#FFE29A") };`
`RagdollRumbleGame.View.cs:103 / void ApplyTheme()`	`case "gym": _bg.color = HzUI.Hex("#FFEBDD"); break;`
`RagdollRumbleGame.View.cs:104 / void ApplyTheme()`	`case "kitchen": _bg.color = HzUI.Hex("#E3FBF1"); break;`
`RagdollRumbleGame.View.cs:105 / void ApplyTheme()`	`case "toybox": _bg.color = HzUI.Hex("#FFE7F3"); break;`
`RagdollRumbleGame.View.cs:106 / void ApplyTheme()`	`case "moon": _bg.color = HzUI.Hex("#A9A6E8"); break;`

RagdollRumbleGame.View.cs:146 / void BuildFrame()	mb.Quad(new Vector2(-40, -H - 4f), new Vector2(40, -H - 4f), new Vector2(40, -H - 0.12f), new Vector2(-40, -H - 0.12f), HzUI.Hex("#B79AD9"), HzUI.Hex("#B79AD9"), HzUI.Hex("#D9C8F0"), HzUI.Hex("#D9C8F0"));
RagdollRumbleGame.View.cs:152 / void BuildFrame()	mb.Rect(new Vector2(s * (W + 0.28f), 0), new Vector2(0.46f, H * 2 + 0.1f), 0, HzUI.Hex("#FFB8D2"));
RagdollRumbleGame.View.cs:156 / void BuildFrame()	mb.Rect(new Vector2(0, H + 0.24f), new Vector2(W * 2 + 1.0f, 0.34f), 0, HzUI.Hex("#FFE29A"));
RagdollRumbleGame.View.cs:230 / void BuildPropView(RrSim.Prop pr)	mb.Rect(Vector2.zero, sz - new Vector2(0.22f, 0.22f), 0, HzUI.Hex("#FFB070"));
RagdollRumbleGame.View.cs:241 / void BuildPropView(RrSim.Prop pr)	var plum = RrArt.Plum; var body = HzUI.Hex("#5C6B7A"); var hi = HzUI.Hex("#8796A6");
RagdollRumbleGame.View.cs:257 / void BuildPropView(RrSim.Prop pr)	mb.Circle(Vector2.zero, r, HzUI.Hex("#2F6F80"), 28);
RagdollRumbleGame.View.cs:258 / void BuildPropView(RrSim.Prop pr)	mb.Circle(new Vector2(-r * 0.3f, r * 0.35f), r * 0.28f, HzUI.Hex("#4E93A5"), 16);
RagdollRumbleGame.View.cs:268 / void BuildPropView(RrSim.Prop pr)	for (int k = 0; k < 10; k++) mb.Rect(new Vector2(0, (k + 0.5f) / 10f), new Vector2(k % 2 == 0 ? 0.13f : 0.1f, 0.1f), 0, k % 2 == 0 ? HzUI.Hex("#C98B4F") : HzUI.Hex("#E2B07A"));
RagdollRumbleGame.View.cs:284 / void BuildPropView(RrSim.Prop pr)	for (float x = -w / 2 + 0.4f; x < w / 2 - 0.2f; x += 0.8f) mb.Rect(new Vector2(x, 0), new Vector2(0.3f, 0.3f), 0, HzUI.Hex("#FFE29A"));
RagdollRumbleGame.View.cs:305 / void BuildPropView(RrSim.Prop pr)	mb.Tri(a, m, c, RrArt.Pink); mb.Tri(m, b, c, HzUI.Hex("#FFF4F8"));
RagdollRumbleGame.View.cs:320 / void BuildPropView(RrSim.Prop pr)	mb.Rect(new Vector2(-sz.x / 2 - 0.08f, 0), new Vector2(0.16f, sz.y + 0.1f), 0, HzUI.Hex("#C9B3EC"));
RagdollRumbleGame.View.cs:321 / void BuildPropView(RrSim.Prop pr)	mb.Rect(new Vector2(sz.x / 2 + 0.08f, 0), new Vector2(0.16f, sz.y + 0.1f), 0, HzUI.Hex("#C9B3EC"));
RagdollRumbleGame.View.cs:326 / void BuildPropView(RrSim.Prop pr)	AddMesh(b.transform, "Vis", Cached("bubble", mb => { mb.Ring(Vector2.zero, 0.07f, 0.11f, HzUI.Hex("#D8FFC4"), 12); }), SortZone + 1);

RagdollRumbleGame.View.cs:361 / void BuildPropView(RrSim.Prop pr)	var on1 = Beam("on1", 3, RrArt.Sun, 0.1f); var on2 = Beam("on2", 9, HzUI.Hex("#FFF6B0"), 0.1f);
RagdollRumbleGame.View.cs:387 / void BuildPropView(RrSim.Prop pr)	mb.Circle(Vector2.zero, r * 0.52f, HzUI.Hex("#7FE0D6"), 20);
RagdollRumbleGame.View.cs:400 / void BuildPropView(RrSim.Prop pr)	mb.Rect(new Vector2(0, sz.y * 0.28f), new Vector2(sz.x - 0.2f, 0.14f), 0, HzUI.Hex("#FFB8D2"));
RagdollRumbleGame.View.cs:428 / void BuildPropView(RrSim.Prop pr)	mb.Rect(new Vector2(0, 0.14f), new Vector2(w - 0.2f, 0.06f), 0, HzUI.Hex("#A8F0E8"));
RrArt.cs:11 / class/static declarations	public static readonly Color Cream = HzUI.Hex("#FFF4E0"), Lilac = HzUI.Hex("#E9DDF7"), LilacDark = HzUI.Hex("#BFA7E0"), Pink = HzUI.Hex("#FF6FA5"),
RrArt.cs:12 / class/static declarations	Sun = HzUI.Hex("#FFC93C"), Teal = HzUI.Hex("#2EC4B6"), Tangerine = HzUI.Hex("#FF8A3D"), Plum = HzUI.Hex("#3A2346"), Ink = HzUI.Hex("#3A2346"),
RrArt.cs:13 / class/static declarations	Grape = HzUI.Hex("#8E6CFF"), Mint = HzUI.Hex("#7FD88A"), Goo = HzUI.Hex("#7DDB4A"), GooDark = HzUI.Hex("#3E9A2A"), Steel = HzUI.Hex("#B8C2CC"),
RrArt.cs:14 / class/static declarations	Good = HzUI.Hex("#3DBE6E"), Danger = HzUI.Hex("#FF4D6A"), Gold = HzUI.Hex("#FFC93C"), Sky = HzUI.Hex("#4DA8FF"), KoGrey = HzUI.Hex("#A9A2B3"),
RrArt.cs:15 / class/static declarations	Panel = new Color(0.23f, 0.14f, 0.27f, 0.88f), PanelLight = new Color(1f, 0.97f, 0.92f, 0.96f), Sub = HzUI.Hex("#F3E6FF"), Locked = HzUI.Hex("#5C4A66");
RrData.cs:115 / public bool HasKoHazard()	new RrProfile { id = "buddy", name = "Buddy", blurb = "A standard test Stuffy.", scale = 1f, massMul = 1f, hp = 100, vuln = 1f, body = HzUI.Hex("#FFF1D6"), accent = HzUI.Hex("#FFC93C") },
RrData.cs:116 / public bool HasKoHazard()	new RrProfile { id = "pip", name = "Pip", blurb = "Small, light and fragile. Flies far!", scale = 0.78f, massMul = 0.55f, hp = 60, vuln = 1.15f, body = HzUI.Hex("#B8F0E6"), accent = HzUI.Hex("#2EC4B6") },

`RrData.cs:117 / public bool HasKoHazard()`	`new RrProfile { id = "bruiser", name = "Bruiser", blurb = "Big and heavy with lots of stuffing. Hard to lift!", scale = 1.25f, massMul = 2.4f, hp = 190, vuln = 0.85f, body = HzUI.Hex("#FFB8D2"), accent = HzUI.Hex("#FF6FA5") },`
`RrData.cs:118 / public bool HasKoHazard()`	`new RrProfile { id = "hopper", name = "Hopper", blurb = "Restless! Flops away from danger every few seconds.", scale = 0.95f, massMul = 0.9f, hp = 100, vuln = 1f, behavior = BHopper, body = HzUI.Hex("#D9F59A"), accent = HzUI.Hex("#7DC63B") },`
`RrData.cs:119 / public bool HasKoHazard()`	`new RrProfile { id = "clinger", name = "Clinger", blurb = "Grabs walls and ledges with its paws. Yank it off!", scale = 1f, massMul = 1.1f, hp = 120, vuln = 1f, behavior = BClinger, body = HzUI.Hex("#FFD3A8"), accent = HzUI.Hex("#FF8A3D") },`

CHAPTER 10

Reskin recipe: visible menu and HUD text

Open `Scripts/RagdollRumbleGame.Hud.cs`, find the displayed literal `"Settings"`, and replace its displayed value with `"Options"`. Keep hierarchy names and technical IDs unchanged. In HzUI.Label/Button calls the first string may be an object name; change the displayed text argument. Read the method references below to distinguish construction from later refresh assignments.

Preserve interpolation braces, invariant number formatting, ON/OFF updates and line breaks. Update both initial and refresh paths, portrait and landscape. Exact helper names/unlocks/charges: Scripts/RagdollRumbleGame.cs; help and effect descriptions: Scripts/RagdollRumbleGame.cs. Level names/hints remain separate JSON content. If a title appears in art, Title property or another screen, update those visible uses deliberately without renaming GameId, namespace, resources or save keys.

Hard-coded game strings do not automatically use shared localization. To translate the interface, route every display assignment through your translation table. Dictionaries, word banks and theme pools are independent gameplay assets and need separate validation. Test help, settings, results, tutorial and helper introductions after a fresh launch.

Source	Exact current code
`RagdollRumbleGame.cs:24 / class/static declarations`	`public static readonly string[] ToolName = { "Punch", "Rocket", "Anvil", "Vortex" };`
`RagdollRumbleGame.cs:170 / void ShowIntro()`	`else { Toast($"Level {_playLevel}: {D.name}", RrArt.Sun, 1.3f, ToastY); ToReady(); }`
`RagdollRumbleGame.cs:223 / protected override void Update()`	`else if (Time.time > _loadToastAt) { _loadToastAt = Time.time + 1.4f; Toast("Building a new arena...", RrArt.Sun, 1.3f, ToastY); }`
`RagdollRumbleGame.cs:411 / void OnTool(int b)`	`if (!Unlocked(b)) { HzAudio.Play(Sfx.Invalid, 0.5f); Toast($"{ToolName[b]} unlocks at level {ToolUnlock[b]}", RrArt.Sub, 1.3f, ToastY); return; }`
`RagdollRumbleGame.cs:421 / void OnTool(int b)`	`Toast($"+2 {ToolName[b]}", RrArt.Sun, 1.1f, ToastY);`
`RagdollRumbleGame.Hud.cs:111 / void LayoutPortrait(float rw, float rh)`	`var b = HzUI.Button("Skip", Root, "Skip (Ad)", RrArt.Grape, Color.white, SkipLevel, 38, 30);`
`RagdollRumbleGame.Hud.cs:123 / void BuildLevelPill(RectTransform parent, Vector2 anchor, Vector2 pos, Vector2 size)`	`_levelText = HzUI.Label("T", pill.transform, "", 56, RrArt.Sun);`

RagdollRumbleGame.Hud.cs:126 / void BuildLevelPill(RectTransform parent, Vector2 anchor, Vector2 pos, Vector2 size)	_nameText = HzUI.Label("N", pill.transform, "", 30, RrArt.Sub, TextAnchor.MiddleCenter, false);
RagdollRumbleGame.Hud.cs:138 / void BuildLevelPill(RectTransform parent, Vector2 anchor, Vector2 pos, Vector2 size)	var t = HzUI.Label("T", card.transform, "Stuffing", 34, RrArt.Sub);
RagdollRumbleGame.Hud.cs:161 / void BuildHpRows(RectTransform parent, float w, float h)	var name = HzUI.Label("Name", row, P.name, 26, Color.white, TextAnchor.MiddleLeft, false);
RagdollRumbleGame.Hud.cs:184 / void BuildMovesInner(RectTransform parent, float w, float h)	_movesText = HzUI.Label("Moves", parent, "Moves 0", 44, Color.white);
RagdollRumbleGame.Hud.cs:192 / void BuildMovesInner(RectTransform parent, float w, float h)	_parText = HzUI.Label("Par", parent, "", 26, RrArt.Sub, TextAnchor.MiddleCenter, false);
RagdollRumbleGame.Hud.cs:217 / void BuildTool(RectTransform parent, int b, Vector2 anchor, Vector2 pos, float size)	v.Name = HzUI.Label("Name", bg.transform, ToolName[b], 28, Color.white);
RagdollRumbleGame.Hud.cs:224 / void BuildTool(RectTransform parent, int b, Vector2 anchor, Vector2 pos, float size)	v.BadgeText = HzUI.Label("N", v.Badge.transform, "", (int)(size * 0.24f), Color.white); v.BadgeText.rectTransform.Stretch();
RagdollRumbleGame.Hud.cs:231 / void BuildTool(RectTransform parent, int b, Vector2 anchor, Vector2 pos, float size)	var lt = HzUI.Label("Lv", lk.transform, "Lv " + ToolUnlock[b], (int)(size * 0.2f), RrArt.Sun);
RagdollRumbleGame.Hud.cs:245 / void RefreshTools()	v.BadgeText.text = _guide == b ? "1" : c > 0 ? c.ToString() : "+";
RagdollRumbleGame.Hud.cs:277 / void RefreshHud(bool full)	_levelText.text = $"Level {_playLevel}";
RagdollRumbleGame.Hud.cs:278 / void RefreshHud(bool full)	_nameText.text = D != null ? D.name : "";
RagdollRumbleGame.Hud.cs:290 / void RefreshHud(bool full)	h.Label.text = d.KO ? d.P.name + " - KO!" : d.P.name;
RagdollRumbleGame.Hud.cs:298 / void RefreshHud(bool full)	_movesText.text = "Moves " + moves.ToString();
RagdollRumbleGame.Hud.cs:299 / void RefreshHud(bool full)	_parText.text = "3 stars: " + D.par.ToString() + " or less";

RagdollRumbleGame.Hud.cs:387 / void ShowLevelCard(Action go)	var p = ArtPopup($"Level {_playLevel}", msg, RrArt.Grape, art, 0, ("Rumble!", RrArt.Good, go));
RagdollRumbleGame.Hud.cs:389 / void ShowLevelCard(Action go)	var sub = HzUI.Label("Name", p.Card, D.name, 40, RrArt.Pink);
RagdollRumbleGame.Hud.cs:396 / void UnlockPopup(int b, Action done)	ArtPopup("New: " + ToolName[b] + "!", ToolDesc[b] + $"\nYou get {ToolStart[b]} free!", HzUI.Hex(ToolColor[b]), RrArt.Get(ToolArt[b]), 0, ("Great!", RrArt.Good, done));
RagdollRumbleGame.Hud.cs:405 / void ShowWinPopup(int done, int moves, int stars)	ArtPopup(done == 25 ? "Lab Champion!" : "Knockout!", msg, RrArt.Good, RrArt.Get("ko_badge") ?? RrArt.Get("win_badge"), stars,
RagdollRumbleGame.Hud.cs:416 / void ShowSettingsPopup()	("How to Play", RrArt.Sun, () => HelpPage(0)),
RagdollRumbleGame.Hud.cs:421 / void ShowSettingsPopup()	var p = Popup("Settings", "", RrArt.Grape, list.ToArray());
RagdollRumbleGame.Hud.cs:431 / static string SoundLabel() => HzAudio.Muted ? "Sound: OFF" : "Sound: ON";	static string SoundLabel() => HzAudio.Muted ? "Sound: OFF" : "Sound: ON";
RagdollRumbleGame.Hud.cs:432 / static string MusicLabel() => HzAudio.MusicMuted ? "Music: OFF" : "Music: ON";	static string MusicLabel() => HzAudio.MusicMuted ? "Music: OFF" : "Music: ON";
RagdollRumbleGame.Hud.cs:444 / void HelpPage(int i)	Popup($"How to Play {i + 1}/{HelpPages.Length}", HelpPages[i], RrArt.Sun, (last ? "Got it!" : "Next", RrArt.Good, last ? (Action)null : () => HelpPage(i + 1)));
RagdollRumbleGame.Hud.cs:455 / void BuildGuideLayer()	_tipText = HzUI.Label("T", _tip, "", 36, Color.white, TextAnchor.MiddleCenter, false);
RagdollRumbleGame.Hud.cs:657 / void EndGuide(bool used)	if (used) Toast("Nice! Tools refill with rewarded ads.", RrArt.Sun, 1.6f, ToastY);
RagdollRumbleGame.View.cs:856 / void FloatText(Vector2 world, string text, Color col, int size, float life)	var l = HzUI.Label("Float", Overlay, "", 50, Color.white);

CHAPTER 11

Artwork, fonts, audio and licences

Replace game images under Resources/ragdollrumble through Unity while retaining resource keys, .meta GUIDs, Sprite/UI import settings, alpha, pivots and slicing. Procedural graphics in the art/view sources may need separate colour changes. Replacing one PNG does not change every fallback. Keep third-party font and other licence notices with the corresponding shipped assets.

Effects are under Resources/ragdollrumble/Sfx. Preserve each actual load key and importer settings when replacing sounds. Supplied music: Resources/ragdollrumble/Music/plush_loop.ogg. Preserve that load key and verify loop/compression and the independent Music toggle. Kit UI uses supplied fonts; preserve actual licence notices and verify new glyphs.

CHAPTER 12

Local saves, reset and migration

HzSave wraps PlayerPrefs and flushes local state on focus loss/quit. WebGL persistence uses browser storage tied to the hosting origin. Clearing site data or changing origin starts a new local save; this is not cloud synchronization. Inspect HzSave/PlayerPrefs calls in the actual game source before altering progression, stars, helper charges or tutorial flags.

Verify fresh launch, a completed arena, helper use, sound options, focus loss and restart. Use a dedicated test profile for resets. Keep technical save keys stable during colour/text reskins. If changing schema or keys, implement and test a migration rather than silently discarding existing users' progress.

CHAPTER 13

Advertising integration and configuration

Asset: `Assets/Szekipapa77Ads/Resources/AdsSettings.asset`. Inspect backend, useGoogleTestIds and missingSdkFallback in the publisher-owned Ads configuration. Buyer projects include adapter/integration code and exclude external provider SDK binaries. WebGL and a missing-provider fallback simulate ads; these are not live commercial impressions.

For real Android ads install the supported provider SDK from its official source into your working project, enter your own app/placement IDs and use test ads first. Verify banner/interstitial/rewarded callbacks on supported hardware. For production use your own IDs and disable simulated missing-SDK fallback. Build tooling rejects unsafe production fallback settings. No IAP, remote backend or HLK is included in this publisher edition.

CHAPTER 14

Android build and release

Install Unity 6000.0.62f1 Android modules through Hub. Select Android, run Setup and review application ID `com.szekipapa77.ragdollrumble`, version, scene orientation and signing. The build command writes `Builds/Android/ragdollrumble.apk`. Confirm BuildResult.Succeeded and the actual output file; intermediate report totals are not the final APK size.

Use your own keystore/signing and distribution settings. A test-signed demo is not a signed store release. A provider-free build simulates ads; a provider test-demo does not prove production ad-account configuration. Test real hardware touch, safe areas, focus/audio, save/relaunch and reward callbacks. Keep passwords, keystores and private service credentials out of buyer packages.

CHAPTER 15

WebGL build and hosting

Install matching WebGL support. Run Setup and Build WebGL; output is `Builds/WebGL/ragdollrumble/`. Serve index.html with all loader/framework/data/WASM/StreamingAssets files over HTTP(S), not file://. Match compression, Content-Encoding and MIME types to the host. Outer page sizing and actual scene orientation are different settings.

The publisher's playable demo is hosted on HyperCodeStore. Upload the actual runtime files rather than merely changing a URL. Test a fresh anonymous load, real pointer controls, audio after interaction, saved progress/reload and phone/desktop sizing. Rebuild and rehost when code/assets change. A successful HTTP page alone is not gameplay verification.

CHAPTER 16

Troubleshooting

Compile errors: restore matching packages and the full private Kit/Ads layer; only one game Kit belongs in a project. **Pink/invisible rendering:** run Setup and inspect Graphics and every Quality URP assignment. **No input:** classic input handling must be enabled; check canvas focus and blocking modals. **Missing resources:** restore exact paths, .meta GUIDs and importer settings.

Edits do not appear: stop Play, Save in the level editor, reload the canonical bank and check current saved level. **Text reverts:** update construction and refresh assignments from the exact table. **No live ads:** provider binaries are deliberately excluded; inspect your installed provider and fallback mode. **Scene reload exception:** retain owner-lifetime and generation guards on delayed callbacks. **WebGL fails:** inspect browser Console, compression headers and missing payloads.

CHAPTER 17

Support and final release checklist

Publisher: Szekipapa77. Support: szekipapa77@gmail.com. Include game title, exact Unity version, platform, first Console error and reproduction steps. Never send passwords/tokens.

1. Clean-import the standalone project and native package.
2. Play both scenes and replace a scene near startup.
3. Review tutorial, normal completion/retry and every helper activation plus effect.
4. Validate editor changes and test reskin text/colours in both layouts.
5. Rebuild affected Android/WebGL outputs.
6. Verify complete public APK bytes, actual browser demo, documentation and video links.
7. Keep licences and exclude all internal prompts, planning/capture/build/QA tooling from buyers.